PEDNet: Prior-Guided Multielliptical Frequency Masking and Dual-Domain Fusion for Remote Sensing Semantic Segmentation

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***Abstract*—Semantic segmentation of high-resolution remote sensing imagery (RSIs) remains challenging due to the coexistence of direction-sensitive structures, complex land-cover patterns, and insufficient class discriminability. Existing methods often emphasize either spatial-domain local details or global contextual aggregation, while paying limited attention to the anisotropic frequency characteristics induced by real ground objects, often resulting in the suboptimal learned representations. To address this issue, we propose a prior-guided multielliptical frequency masking and dual-domain fusion network (PEDNet), a unified prior-guided anisotropic spatial–frequency modeling framework for remote sensing semantic segmentation (RSSS). The core idea of PEDNet is to align frequency modeling with the directional geometry and semantic heterogeneity of land-cover objects, rather than relying on fixed isotropic decompositions. Specifically, a multielliptical frequency decomposition module (MEFDM) is introduced to perform anisotropy-aware frequency partitioning in the Fourier domain using learnable elliptical masks, thereby enhancing anisotropic spectral modeling and boundary-detail perception. In addition, a contextualized dual-attention fusion module (CDAFM) and a prior-guided mask adaptation module (PGMAM) are developed to realize coordinated spatial–frequency interaction and scene-adaptive frequency modeling, where CDAFM strengthens cross-domain feature integration through contextualized bidirectional attention and gated fusion, and PGMAM dynamically refines frequency partitioning under semantic prior guidance. Extensive experiments on multiple benchmark datasets demonstrate that PEDNet consistently achieves superior segmentation performance, especially in challenging scenarios involving fine boundaries, anisotropic structures, and densely distributed small objects.**

***Index Terms*—Contextualized dual-attention fusion, multielliptical frequency decomposition, prior-guided mask adaptation, spatial–frequency domain.**

I. Introduction

HIGH-RESOLUTION remote sensing imagery (RSIs) provides a detailed characterization of Earth's surface, serving as a fundamental data source for a wide range of geospatial analysis tasks and environmental management processes, including large-scale urban system analysis [1], ecological and environmental assessment [2], and land-use investigation. Reference [3] among various remote sensing interpretation tasks, semantic segmentation assigns semantic labels for each pixel to ground objects [4], thereby enabling a structured understanding of complex geospatial scenes. Nevertheless, accurate semantic understanding of large-scale RSIs remains challenging. Complex texture patterns introduced by heterogeneous objects [5] frequently blur object boundaries, while long-range contextual dependencies caused by spectral variability [6] and multiscale spatial layouts [7] further complicate feature discrimination. These factors jointly hinder precise boundary delineation, reliable small object detection, and discriminative semantic representations of class learning in high-resolution RSIs.

Although traditional machine learning (ML) approaches, such as logistic regression [8], clustering [9], and support vector machines (SVMs) [10], together with handcrafted feature descriptors based on texture [11] and color [12], [13], have achieved notable progress; their inherent limitations in modeling complex spatial structures restrict their effectiveness in large-scale and fine-grained scene understanding. In contrast, convolutional neural networks (CNNs)-based methods have become the dominant paradigm for RSSS, owing to their strong capability for hierarchical and discriminative feature representation learning.

Classical CNN architectures, including FCN [14], UNet [15], and DeepLab seris [16], [17], [18], typically rely on hierarchical CNN-based encoders combined with multiscale skip connections to gradually fuse contextual information across different levels. However, these architectures have not been readily transferred to RSIs, where scenes are characterized by strong spatial heterogeneity, irregular object distributions, and significant scale variations. To better accommodate these domain-specific challenges, a wide range of CNN-based RSSS methods have been developed, incorporating strategies such as multilevel feature aggregation [19], boundary-aware modeling [20], and attention-driven representation enhancement [21].

Despite their effectiveness in RSSS, CNN-based approaches' locality constraints the modeling of long-range dependencies, often producing fragmented predictions [22] and blurred object boundaries [23] when dealing with complex aerial scenes. To overcome these limitations, Transformer [24] architectures were later introduced from the natural image domain, where global self-attention enables the modeling of global spatial dependencies beyond local

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receptive fields. The vision transformer (ViT) [25] and its subsequent variants [26], [27], [28] further treat image understanding as a sequence modeling task, enabling enhanced global reasoning and flexible contextual aggregation. With regard to RSSS, Transformer-based models have been widely explored and have demonstrated clear advantages in handling spatial heterogeneity [29], large object scale variations [30], and complex spectral and spatial interactions [31]. However, the computational cost of self-attention grows rapidly with feature resolution, which remains a major bottleneck for large-scale RSIs.

Although existing CNN-based and Transformer-based segmentation methods are able to capture global contextual information, they still face several limitations when applied to high-resolution RSIs. First, most of these models focus mainly on spatial features and pay little attention to frequency information or directional cues. As a result, they struggle to describe repetitive textures and clear structural edges, which are especially important for separating man-made objects from surrounding natural areas. Second, spatial features and frequency-related information are often handled independently, making it difficult to effectively combine global semantic context with fine local structures. This separation can weaken small-object recognition and lead to broken or inconsistent structures in scenes with dense object distributions. Finally, many existing approaches do not fully exploit domain-specific prior knowledge, such as typical land-cover patterns or terrain characteristics. Without these priors, models lack adaptive feature adjustment, which in turn limits their robustness and discriminative ability across different geographic regions.

To tackle these problems, we argue that a more effective representation for RSIs should jointly model spatial and frequency information, rather than treating them as two isolated domains, while also leveraging semantic priors related to ground objects. In particular, many RSIs contain direction-sensitive structures such as roads, building boundaries, and elongated land parcels. These structures are not only spatially anisotropic but also tend to induce noncircular and direction-aware energy distributions in the Fourier domain. This suggests that frequency decomposition for RSSS should not rely solely on isotropic frequency partitioning but should instead account for anisotropic frequency characteristics. Motivated by this observation, we propose a spatial–frequency collaborative framework, termed prior-guided multielliptical frequency masking and dual-domain fusion network (PEDNet) for RSSS. PEDNet integrates anisotropy-aware frequency decomposition, spatial–frequency interaction, and semantic priors into a unified framework, allowing these components to work together in a more coordinated and task-relevant manner.

1) We propose *a unified prior-guided anisotropic spatial–frequency modeling framework for RSSS, namely PEDNet*. The central idea is to align frequency modeling with directional geometry and semantic heterogeneity of land-cover objects rather than relying on fixed isotropic decompositions, providing a coherent basis for decomposition, fusion, and adaptation within a single framework.
2) We develop *a multielliptical frequency decomposition module* (MEFDM) to perform anisotropy-aware frequency partitioning in the Fourier domain. By introducing learnable elliptical masks with directional parameters, MEFDM enables the decomposition process to better conform to direction-sensitive structures in RSIs, capturing anisotropic patterns and boundary-related high-frequency details more effectively.
3) We further develop *a contextualized dual-attention fusion module* (CDAFM) and *a prior-guided mask adaptation module* (PGMAM) to realize coordinated spatial–frequency interaction and scene-adaptive frequency modeling. Specifically, CDAFM enhances cross-domain feature integration through contextualized bidirectional attention and gated fusion, while PGMAM dynamically adjusts frequency partitioning under semantic prior guidance.

The remainder of this article is organized as follows. Section II reviews related works on CNN-based methods, Transformer-based methods, and frequency-domain techniques for RSSS. Section III presents the methodology of the proposed PEDNet framework in detail. Section IV describes the experimental settings, including the datasets, evaluation metrics, and implementation details. Sections V and VI report the ablation studies of proposed modules and comparative experiments against classical and recent state-of-the-art (SOTA) approaches, respectively. Finally, Section VII concludes this article.

II. Related Works

A. CNN-Based Methods for RSSS

Deep convolutional architectures have established the foundation paradigm for RSSS, providing hierarchical feature extraction and multiscale spatial representation. Lv et al. [32] proposed a symmetric encodingCdecoding architecture that performs hierarchical feature fusion during both representation extraction and reconstruction. Wang et al. [33] proposed MFANet, which incorporated a hierarchical feature integration module and a pyramid-based representation scheme to generate fine-grained feature maps. Bai et al. [34] extended the FCN framework with attention-guided feature modulation across different receptive fields and a nonlocal filtering operation, enabling richer contextual feature interaction. Liu et al. [35] further explored an eigenvalue-preserving strategy to facilitate robust building extraction across varying spatial resolutions in RSIs. Generally, hierarchical feature fusion has been widely explored to combine semantic information from different layers, as it helps produce more consistent structures in complex scenes. However, many of these fusion-focused designs mainly prioritize global consistency, often overlooking high-frequency details that are important for accurately describing fine structures.

To alleviate this limitation, several studies have explored the idea of explicitly modeling object boundaries within segmentation networks. For example, Pan et al. [36] used traditional edge detection and morphological operations to

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generate edge-aware supervision features. Chong et al. [37] introduced an information interaction strategy to refine object contours, particularly for small-scale targets. More recently, Miao et al. [38] improved edge sensitivity by introducing class activation guided boundary enhancement, yielding clear benefits for small object segmentation.

Although these methods have improved boundary quality to some extent, CNN-based architectures still struggle to model interactions across distant spatial regions and to capture the structured spectral patterns commonly found in high-resolution RSIs. As a result, models that rely purely on spatial convolutions often find it difficult to distinguish fine-grained textures and repetitive patterns, which are important cues for separating man-made structures from natural surfaces when viewed from a frequency perspective.

B. Transformer-Based Methods for RSSS

Transformer-based architectures have been increasingly adopted in RSSS due to their ability to establish global contextual interactions beyond the locality constraints of convolutional operations. Early studies explored Transformer designs tailored for RSIs by incorporating semantic guidance or hierarchical attention mechanisms. For instance, Meng et al. [39] introduced a class-guided Swin Transformer that embeds category-aware cues into a Transformer-based encoding–decoding paradigm. Liu et al. [40] investigated a global–local Transformer segmentation framework in which both encoding and decoding stages are jointly optimized to promote consistent feature representation. Despite these advances, purely Transformer-driven models often encounter difficulties in preserving fine structural details, as the absence of convolutional inductive biases can weaken sensitivity to local boundaries and textural patterns. To compensate for this limitation, hybrid architectures that combine operations with Transformer modules have been actively explored. Xiao et al. [41] proposed a hybrid framework to integrate local feature extraction with global contextual reasoning. Wang et al. [42] designed a UNet-style architecture equipped with a Transformer-based decoder for efficient urban scene segmentation, while Chen et al. [43] employed a ResNet-50 encoder coupled with a Transformer decoder to jointly aggregate local representations and nonlocal contextual information.

Beyond terrestrial Earth observation scenarios, Transformer-based segmentation frameworks have also been investigated for planetary surface analysis, such as MarsFormer. While most remote sensing studies focus on Earth observation, the underlying principles of spectral–spatial analysis and semantic segmentation remain equally applicable to extraterrestrial environments such as Mars. For example, MarsFormer [44] adopted an encoder–decoder configuration augmented with feature enhancement and window-based Transformer blocks, where hierarchical representations are progressively constructed via a Mix Transformer. However, planetary exploration missions such as Mars reconnaissance impose strict constraints on on-board computational resources, rendering large-scale Transformer models impractical for direct deployment. To balance segmentation accuracy with computational efficiency under such constraints, Dong et al. [45] introduced a distillation-based approach that facilitates knowledge exchange between convolutional networks and Transformer models. Nevertheless, most existing Transformer variants predominantly focus on spatial correlations, while largely overlooking spectral regularities that encode repetitive or directionally structured patterns in RSIs. In addition, spatial and spectral representations are typically processed in a decoupled manner, which restricts effective cross-domain interaction and limits semantic coherence across different spatial scales.

C. Frequency-Domain and Hybrid Spatial–Spectral Modeling for RSSS

In parallel, frequency-aware modeling strategies have attracted growing interest for their ability to expose structural regularities and textural smoothness that are not readily captured in the spatial domain. To this end, a variety of Fourier- and wavelet-enhanced networks have been explored to enrich spectral representations within deep segmentation models.

Liu et al. [46] adopted an adaptive Fourier convolution layer to model long-range dependencies for extracting long and continuous road structures. Cao et al. [47] introduced GF-ResFormer for global–local modeling and edge feature enhancement. Zhang et al. [48] designed a frequency-domain guided feature coupling mechanism through dilation group convolution and fast Fourier transform (FFT). Yang et al. [49] introduced a frequency-driven awareness generator mechanism that derives semantic-aware activation maps from spectral representations.

More recently, several frequency-domain approaches have been developed specifically for remote sensing semantic segmentation (RSSS) from different perspectives. Ma et al. [50] proposed a decomposition-based unsupervised domain adaptation framework, where high- and low-frequency decomposition is introduced to promote domain-invariant representation learning for cross-domain remote sensing segmentation. Chen et al. [51] explored frequency-domain fusion for multimodal RSSS, demonstrating that frequency-aware multimodal interaction can enhance complementary information aggregation across heterogeneous modalities. Li et al. [52] designed a frequency domain-enhanced spectral–spatial fusion transformer to jointly characterize spatial layouts, structural boundaries, and heterogeneous textures from the frequency perspective. In addition, Li et al. [53] proposed a frequency-guided denoising network to enhance frequency-sensitive representations and improve feature fusion consistency for high-resolution RSSS.

Beyond Fourier-based formulations, wavelet-integrated architectures have also been investigated for multiband frequency analysis. Wei et al. [54] designed a hierarchical wavelet decomposition framework that aggregates frequency features at different resolution levels to improve category separability among spatially similar land-cover classes.

While these methods introduce valuable spectral priors and demonstrate the effectiveness of frequency-domain modeling in RSSS, several limitations still remain. First, many existing frequency-domain approaches rely on fixed filters, predefined decomposition strategies, or manually designed frequency

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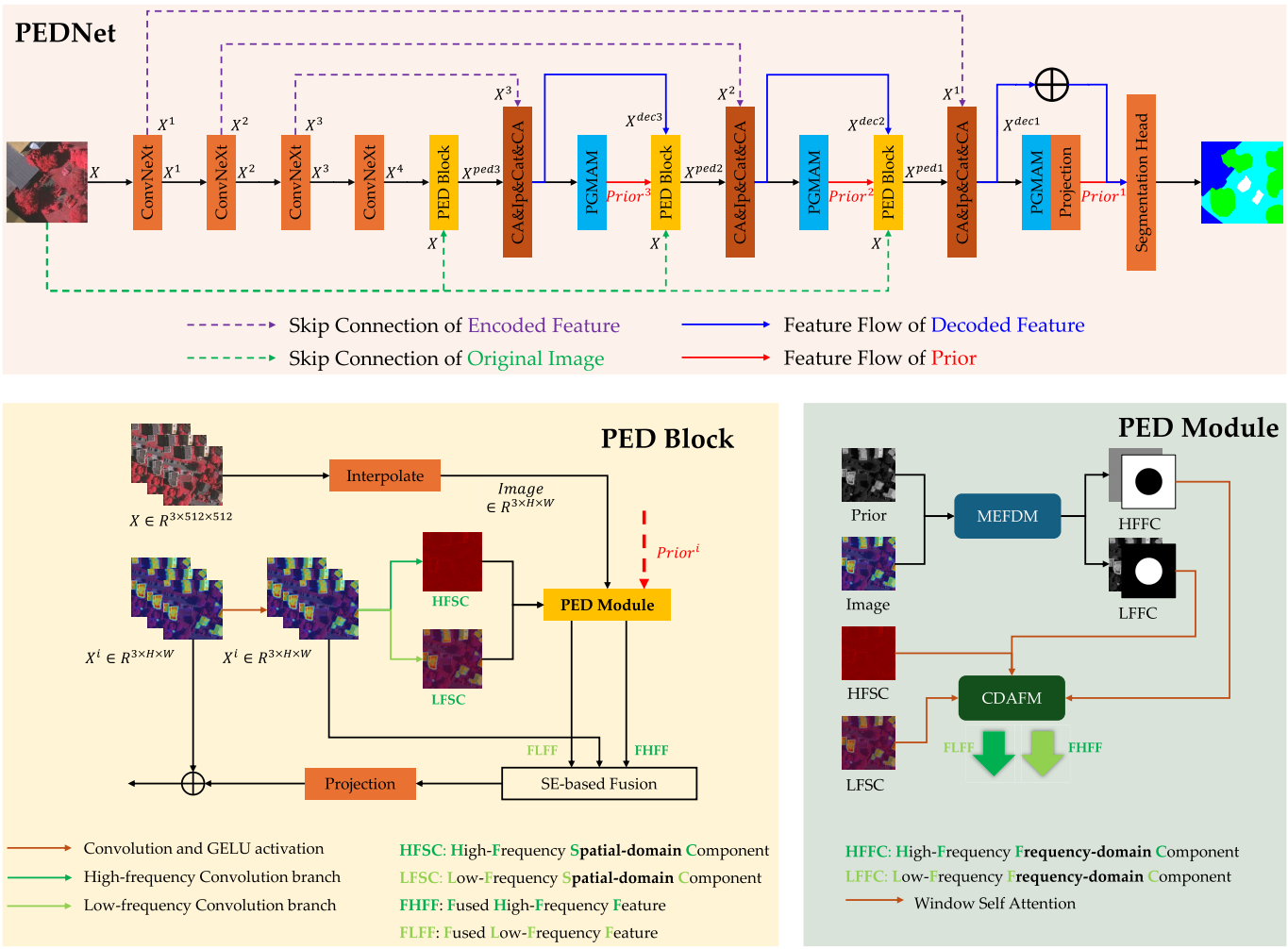


Fig. 1. Structure of the proposed PEDNet. Among the figures, different colors, arrow shapes, and solid or dashed styles represent the flow directions of different features. Specifically, the PED Block consists of the PED Module together with several other necessary components, while the PED Module is jointly composed of MEFDM, CDAFM, and several other essential attention computation components.

masks, which may limit their adaptability to the semantic variability of diverse land-cover types. Second, although recent studies have explored frequency decomposition, multimodal frequency fusion, spectral–spatial enhancement, and frequency-guided denoising, distributional priors inherent to geospatial data, such as terrain morphology and land-use regularity, are still not sufficiently exploited during frequency modulation. Third, the lack of explicit semantic guidance in frequency adaptation may lead to uneven feature enhancement and reduced robustness when generalizing across heterogeneous remote sensing scenes. In contrast, our method aims to incorporate prior-guided frequency-domain modeling with hybrid spatial-spectral interaction, thereby enabling more adaptive isolation and fusion of informative spectral components for RSSS.

III. Proposed PEDNet Framework

This section introduces the overall architecture and concrete design of PEDNet. ConvNeXt-Tiny is adopted as the encoder, while the decoder is constructed to jointly realize anisotropy-aware frequency decomposition, contextualized spatial–frequency interaction, and semantic prior-guided mask adaptation in a coordinated manner. Section III-A provides an overview of the complete framework. Section III-B further explains the motivation for elliptical masking to clarify the design rationale of anisotropic frequency modeling strategy. Sections III-C–III-E detail the three key components of PEDNet, namely MEFDM, CDAFM, and PGMAM.

A. Overview of the Proposed PEDNet

Fig. 1 outlines the framework of the proposed PEDNet, which is organized around an encoding–decoding framework. In the encoder part, ConvNeXt-Tiny is employed as the backbone to achieve a compact yet efficient design, maintaining a reasonable parameter scale while ensuring strong representational capacity. Given the central role of the backbone, the proposed PEDNet is divided into four hierarchical stages. For an input image $X \in \mathbb{R}^{H\times W\times 3}$, the encoder constructs a four-level feature pyramid with channel dimensions increasing while spatial resolutions decrease. The resulting feature tensors are denoted as $X^l \in \mathbb{R}^{H/(2^{l+1})\times W/(2^{l+1})\times 2^{l-1}C}$ for $l \in 1, 2, 3, 4$. Throughout this work, we set $H = W = 512$ and $C = 96$.

As shown in the Fig. 1, the decoder of our framework is designed in a hierarchical top-down manner, progressively

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integrating multiscale features from the encoder. Starting from the deepest representation X^4, the decoder successively upsamples and merges features with shallower encoder inputs (X^3, X^2, and X^1), while channel reduction is applied to maintain a compact representation at each stage. At each decoding level, MEFDM and CDAFM are employed sequentially, which adaptively refine the fused feature with the aid of priors generated by PGMAM from the preceding stage. These priors are progressively propagated from deeper to shallower layers, thereby enhancing the spatial and frequency representations across levels. Finally, in the last decoding stage, an additional prior is explicitly injected into the feature map via residual projection, serving as a terminal calibration. Finally, a dense prediction is obtained from the segmentation head and spatially projected back to the original image size.

B. Motivation for Elliptical Masking

The design of PEDNet is motivated by a simple yet practically important observation that the Fourier spectra of RSIs are often not ideally described by isotropic circular support. Compared with natural images, RSIs contain more direction-sensitive structures such as roads and elongated land parcels. These structures tend to induce noncircular spectral responses, making purely isotropic spectral partitioning less suitable for frequency decomposition. In this context, circular masking is inherently limited because it allocates the same spectral bandwidth to all directions. By contrast, elliptical masking provides anisotropic spectral support through different major and minor axes and therefore is capable of better adapted to direction-aware spectral patterns. Since circular masking is a special case of elliptical masking when the two axes are equal, elliptical masking can be viewed as a compact and natural generalization of isotropic spectral partitioning rather than an arbitrary shape choice.

To provide direct evidence for this motivation, we perform an equal-area spectral coverage analysis on several representative remote sensing datasets, where circular and elliptical masks are compared under the same spectral support area. For fair comparison, we first define a circular spectral support whose radius is set to 20% of the shorter side of the spectrum. The elliptical mask is then constructed under the equal-area constraint, so that both masks have the same support area and differ only in their geometric allocation across directions. Therefore, the reported circular and elliptical coverage values are not predefined energy percentages, but the proportions of Fourier magnitude captured by the two mask shapes under the same spectral support budget.

As shown in Table I, elliptical masks outperform circular masks on the majority of samples in most datasets. In particular, the ellipse better ratios reach 52.2% on Vaihingen, 59.1% on Potsdam, 77.1% on LoveDA-rural, and 78.0% on LoveDA-urban, while the corresponding average coverage gains remain consistently positive. These results indicate that the advantage of elliptical masking is not restricted to a few special cases. More importantly, elliptical masking provides more suitable geometric support for broadly distributed noncircular spectral patterns in RSIs.

TABLE I

EQUAL-AREA SPECTRAL COVERAGE ANALYSIS OF CIRCULAR AND ELLIPTICAL MASKS ON REPRESENTATIVE REMOTE SENSING DATASETS. FOR FAIR COMPARISON, THE CIRCULAR AND ELLIPTICAL MASKS ARE CONSTRUCTED WITH THE SAME SUPPORT AREA, AND THEIR COVERAGE VALUES ARE COMPUTED AS THE PROPORTIONS OF FOURIER MAGNITUDE CAPTURED WITHIN THE CORRESPONDING MASK REGIONS. "GAIN" DENOTES THE AVERAGE DIFFERENCE BETWEEN ELLIPTICAL COVERAGE AND CIRCULAR COVERAGE OVER ALL SAMPLES. "ELLIPSE BETTER RATIO" DENOTES THE PERCENTAGE OF SAMPLES FOR WHICH THE ELLIPTICAL MASK ACHIEVES HIGHER COVERAGE THAN THE CIRCULAR MASK. A HIGHER ELLIPSE BETTER RATIO INDICATES THAT ELLIPTICAL MASKING MORE FREQUENTLY PROVIDES BETTER GEOMETRIC SUPPORT FOR THE SPECTRAL DISTRIBUTION THAN CIRCULAR MASKING UNDER THE SAME AREA BUDGET

Dataset	Circular Coverage	Elliptical Coverage	Gain	**Ellipse Better Ratio**
Vaihingen	0.576266	0.576299	0.000033	**52.2%**
Potsdam	0.578737	0.579135	0.000398	**59.1%**
LoveDA-rural	0.719427	0.720051	0.000623	**77.1%**
LoveDA-urban	0.555731	0.556605	0.000874	**78.0%**

Fig. 2 further shows representative examples of the Fourier spectra of RSIs. One can observe that the spectral energy often spreads preferentially along certain directions rather than forming an ideal circular pattern. Under the equal-area constraint, elliptical masks better align with such noncircular spectral distributions than circular ones. Motivated by these observations, we adopt multielliptical spectral partitioning as the basis of PEDNet and further build dual-domain interaction and prior-guided adaptation upon it.

C. Multielliptical Frequency Decomposition Module

As shown in Fig. 3, given an input image optionally with a prior map, MEFDM learns multiple axis-aligned multielliptical masks in the frequency plane and uses it to adaptively split features into frequency components, which are later refined and fused with spatial features. Let the raw image be denoted as $X \in \mathbb{R}^{B\times 3\times H\times W}$ and an optional prior be denoted as $P \in \mathbb{R}^{B\times p\times H\times W}$. At the current stage, the feature dimensionality is C, and the number of elliptical masks is K. We first form a shallow frequency feature denoted as Z

$$Z = \mathrm{Conv}_{3\times 3}([X, P]) \in \mathbb{R}^{B\times C\times H\times W} \tag{1}$$

where $Conv_{3\times 3}$ denotes the convolution operation under the kernel size of 3×3 and the prior P will be introduced in Section III-E.

To obtain the parameters of multiple elliptical masks, we design a prediction head consisting of a global average pooling layer followed by two 1×1 convolutions with a GELU activation. The final projection outputs 4K normalized values, corresponding to the center coordinates and the lengths of the two semi-axes of each ellipse, with a Sigmoid function constraining them to the range $[0, 1]$ as follows:

$$\Theta = \sigma(W_2\phi(W_1(GAP(z)))) \in \mathbb{R}^{B\times 4K\times 1\times 1} \tag{2}$$

where σ denotes the Sigmoid function, ϕ denotes the GELU function, W_1 and W_2 denote two 1×1 convolutions, and GAP denotes global average pooling. We broadcast the predicted ellipse parameters across all channels by reshaping and repeating the tensor, ensuring that each channel shares the same set

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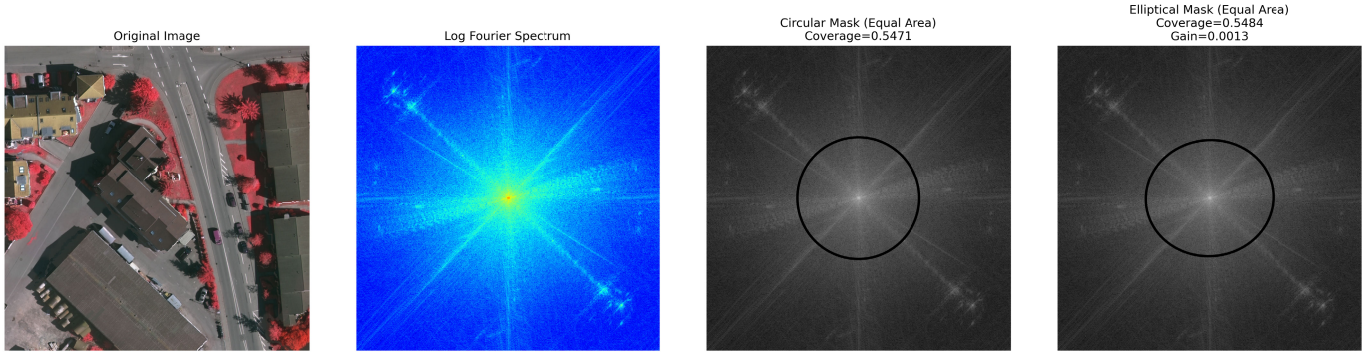


Fig. 2. Visualization of the motivation for elliptical spectral masking. (Left to right) Original image, its log Fourier spectrum, the equal-area circular mask, and the equal-area elliptical mask, together with the corresponding coverage values. It can be observed that the spectral energy of RSIs often deviates from ideal circular symmetry, and the elliptical mask provides higher coverage under the same support area, which supports the use of elliptical spectral partitioning.

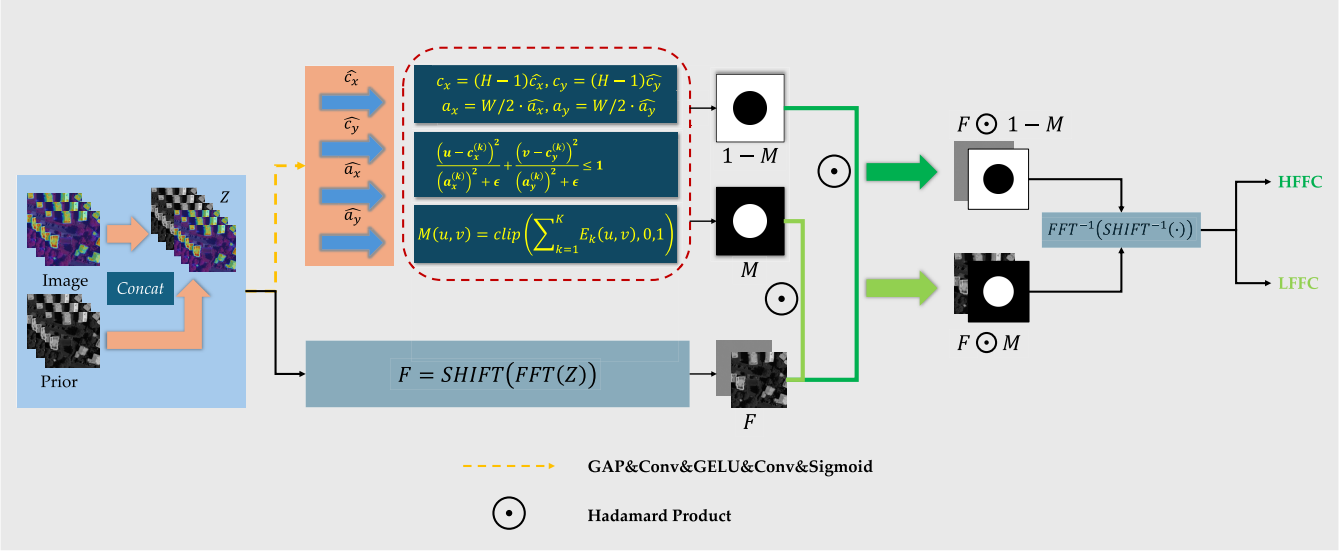


Fig. 3. Structure of the proposed MEFDM. Following the pathways illustrated in the figure, the feature map and the prior map undergo a series of transformations to successively generate the parameters of multiple ellipses, the final frequency-domain mask, and the Fourier-transformed frequency-domain components. After element-wise multiplication, an inverse transform is applied to project the result back into the original feature space.

of K ellipses defined by four parameters each. Therefore, we obtain each ellipse's parameters as

$$\theta_{B,C,K} = \left(\hat{c}_x, \hat{c}_y, \hat{a}_x, \hat{a}_y\right) \in (0,1)^4 \tag{3}$$

and rescale them to the pixel scale according to the input spatial size

$$c_y = (H-1)\,\hat{c}_y, c_x = (W-1)\,\hat{c}_x, a_y = \frac{H}{2}\hat{a}_y, a_x = \frac{W}{2}\hat{a}_x. \tag{4}$$

For each spatial frequency location (u, v), the kth ellipse indicator is

$$\mathbb{E}_k(u,v) = 1\left(\frac{\left(u - c_y^{(k)}\right)^2}{\left(a_y^{(k)}\right)^2 + \varepsilon} + \frac{\left(v - c_x^{(k)}\right)^2}{\left(a_x^{(k)}\right)^2 + \varepsilon} \le 1\right). \tag{5}$$

We obtain the final mask by taking the union of all ellipses and clipping it at 1, i.e.,

$$M(u,v) = \text{clip}\left(\sum_{k=1}^{K} \mathbb{E}_k(u,v), 0, 1\right). \tag{6}$$

The mask is broadcast per channel: $M \in \{0,1\}^{B\times C\times H\times W}$.

To perform the frequency separation, we further compute the 2-D FFT of z with centering

$$F = \text{shift}\left(\text{FFT}(z)\right) \tag{7}$$

where FFT$(\cdot)$ represents the FFT that converts the spatial-domain feature z into its frequency-domain representation. shift$(\cdot)$ denotes a spectrum-centering operation, which rearranges the frequency components by moving the zero-frequency component to the center of the spectrum.

Using M as the high-low frequency channel divider, we obtain

$$f_{low}^{temp} = F \odot M, f_{high}^{temp} = F \odot (\mathbf{1} - M) \tag{8}$$

where $\mathbf{1}$ is an all-ones matrix. It will be obtained after a following complete inversion operation as

$$\begin{aligned} f'_{low} &= \left|\text{IFFT}\left(\text{ishift}\left(f_{low}^{temp}\right)\right)\right| \\ f'_{high} &= \left|\text{IFFT}\left(\text{ishift}\left(f_{high}^{temp}\right)\right)\right| \end{aligned} \tag{9}$$

where ishift$(\cdot)$ denotes the inverse frequency-shift operation, which restores the shifted spectrum to the original frequency layout before the inverse Fourier transform. IFFT$(\cdot)$ denotes

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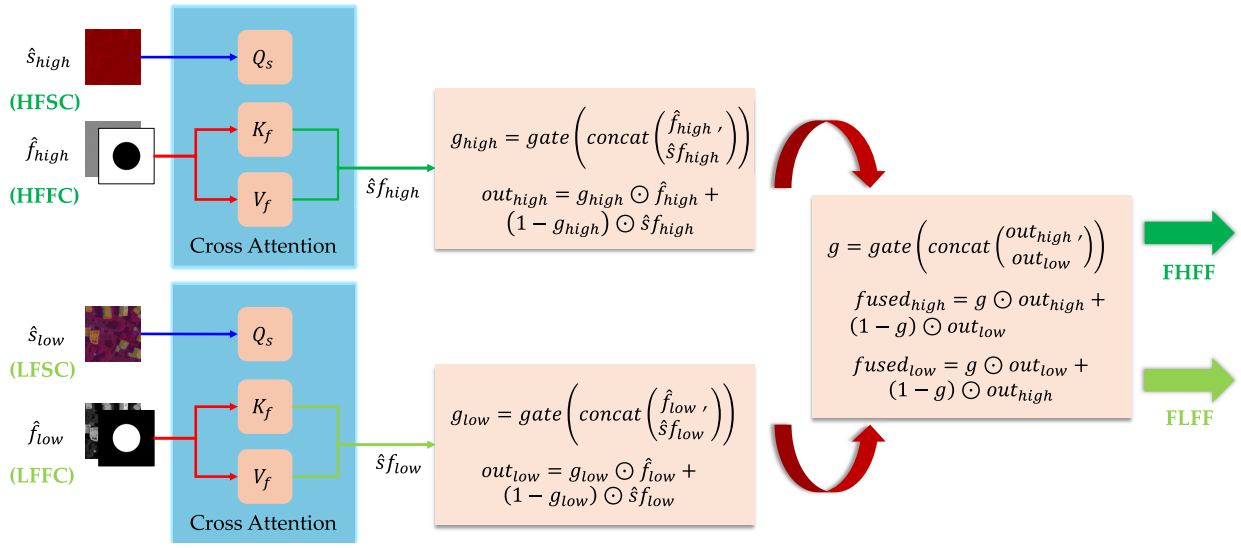


Fig. 4. Structure of the proposed CDAFM. The high- and low-frequency components in both the spatial and frequency domains are further processed by window-based self-attention, cross-attention, and gated fusion mechanisms, thereby generating the final output components.

the inverse FFT, which converts the masked frequency-domain features back to the spatial domain. The magnitude operation $|\cdot|$ is applied to obtain real-valued spatial feature responses from the complex-valued inverse Fourier outputs.

Finally, 1×1 convolutions reduce both branches to $C/2$ channels for subsequent attention and cross-domain fusion

$$f_{low} = Conv_{1\times1}\left(f'_{low}\right) \in \mathbb{R}^{B\times C/2\times H\times W}$$
$$f_{high} = Conv_{1\times1}\left(f'_{high}\right) \in \mathbb{R}^{B\times C/2\times H\times W}. \quad (10)$$

The method described above operates by adaptively partitioning the frequency spectrum through a set of learnable elliptical masks, whose centers, scales, and orientations are jointly optimized to align with the intrinsic structural patterns of the input. This mechanism enables the model to perform anisotropic and multiregion frequency selection, thereby decomposing the feature representation into low- and high-frequency components in a content-aware manner.

The distinctive advantage of this design lies in its ability to transcend the limitations of conventional isotropic masking strategies. By allowing flexible, direction-sensitive, and multicenter spectral filtering, the proposed approach can better capture heterogeneous and multiscale structures commonly observed in complex scenes. As a result, it achieves a more faithful separation of structural information and fine-grained details, leading to enhanced feature expressiveness and improved robustness in dense prediction tasks.

D. Contextualized Dual-Attention Fusion Module

The preceding section describes the acquisition process of frequency-domain features. To further fuse spatial and frequency-domain representations, we extract spatial high- and low-frequency features from the input image, as illustrated in Fig. 4. First, a 7×7 depthwise convolution is employed to extract low-frequency components by capturing broad structural information. Second, the high-frequency components are obtained as residual features by subtracting the low-frequency component from the input. Third, a 3×3 depthwise convolution is used to enhance these high-frequency details. Finally, two 1×1 convolutions project both components to half of the original channels, producing complementary feature representations for the subsequent attention mechanism: $s_{low}, s_{high} \in \mathbb{R}^{B\times C/2\times H\times W}$.

The frequency-domain branch features f_{low}, f_{high} and the spatial-domain branch features s_{high}, s_{low} are then processed by multihead self-attention mechanisms. Taking the feature s_{high} as an example, it is first reshaped into a token sequence with length $N = H \times W$

$$Z \in \mathbb{R}^{B\times \frac{C}{2}\times N}$$
$$Q = ZW_Q, K = ZW_K, V = ZW_V. \quad (11)$$

For each attention head in the multihead attention mechanism, the self-attention operation is formulated as

$$Attn\,(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_h}}\right)V. \quad (12)$$

The outputs of multiple heads are concatenated and then projected back to the feature map dimensions. The features f_{low}, s_{high}, and s_{low} undergo the same operation, yielding four corresponding features after self-attention feature extraction

$$\tilde{f}_{high} = SA_f\left(f_{high}\right), \tilde{f}_{low} = SA_f\left(f_{low}\right)$$
$$\tilde{s}_{high} = SA_s\left(s_{high}\right), \tilde{s}_{low} = SA_s\left(s_{low}\right). \quad (13)$$

Afterward, the frequency-domain and spatial-domain features are fed into a mutual cross-attention mechanism to obtain cross-domain representations. Taking $\tilde{f}_{high}$ and $\tilde{s}_{high}$ as an example, when $\tilde{s}_{high}$ serves as the *Query*, the cross-attention between $\tilde{s}_{high}$ and $\tilde{f}_{high}$ generates $\tilde{sf}_{high}$. Subsequently, $\tilde{f}_{high}$ and $\tilde{sf}_{high}$ are concatenated, and a gating mechanism is applied

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to the combined feature to obtain frequency-domain features enhanced with spatial-domain details

$$\begin{aligned} \hat{f}s_{high} &= crossattn\left(\hat{f}_{high}, \hat{s}_{high}\right) \\ \hat{s}f_{high} &= crossattn\left(\hat{s}_{high}, \hat{f}_{high}\right) \\ g_{high} &= gate\left(concat\left(\hat{f}_{high}, \hat{s}f_{high}\right)\right) \\ out_{high} &= g_{high} \odot \hat{f}_{high} + \left(\mathbf{1} - g_{high}\right) \odot \hat{s}f_{high}. \end{aligned} \tag{14}$$

The same process is applied to the features $\tilde{f}_{low}$ and $\tilde{s}_{low}$

$$\begin{aligned} \hat{f}s_{low} &= crossattn\left(\hat{f}_{low}, \hat{s}_{low}\right) \\ \hat{s}f_{low} &= crossattn\left(\hat{s}_{low}, \hat{f}_{low}\right) \\ g_{low} &= gate\left(concat\left(\hat{f}_{low}, \hat{s}f_{low}\right)\right) \\ out_{low} &= g_{low} \odot \hat{f}_{low} + (\mathbf{1} - g_{low}) \odot \hat{s}f_{low}. \end{aligned} \tag{15}$$

Both out_{high} and out_{low} are frequency-domain features augmented with spatial-domain information. To fuse these two features, another gating mechanism is performed

$$\begin{aligned} g &= gate\left(concat\left(out_{high}, out_{low}\right)\right) \\ fused_{high} &= g \odot out_{high} + (\mathbf{1} - g) \odot out_{low} \\ fused_{low} &= g \odot out_{low} + (\mathbf{1} - g) \odot out_{high}. \end{aligned} \tag{16}$$

Finally, a squeeze-and-excitation mechanism is employed to fuse $fused_{high}$ and $fused_{low}$. After adding the fused feature to the initial input feature map, the output of CDAFM is denoted as

$$\begin{aligned} X_{output-CDAFM} = X_{input-CDAFM} \\ + CDAFM\left(X_{input-CDAFM}\right). \end{aligned} \tag{17}$$

E. Prior-Guided Masking-Adaption Module

To alleviate the ambiguity of frequency-domain decomposition in complex remote sensing scenes, we introduce a PGMAM. Here, the term *prior* does not refer to external prior knowledge, handcrafted geographic rules, or predefined category prototypes. Instead, it denotes an *internally induced semantic prior*, implemented as a single-channel normalized spatial bias map distilled from decoder features and injected before subsequent frequency decomposition and final prediction refinement. In this sense, the prior serves as a coarse category-aware preference for downstream frequency decomposition and final prediction refinement.

Given the decoder feature $U^{(l)} \in \mathbb{R}^{B\times C_l\times H_l\times W_l}$ at level l, we generate a single-channel semantic prior map as

$$P^{(l)} = \sigma\left(W^{(l)}_{prior} * U^{(l)}\right) \tag{18}$$

where $W^{(l)}_{prior}$ denotes a 1×1 convolution, $*$ denotes convolution, and $\sigma(\ldots)$ is the Sigmoid function. Accordingly, $P^{(l)} \in [0,1]^{B\times 1\times H_l\times W_l}$ is a normalized spatial guidance map. Since P^l compresses decoder semantics into a low-dimensional spatial bias while discarding most channel-wise representational redundancy, it is more appropriately interpreted as a learned semantic prior than as a full contextual feature tensor.

The generated prior is injected into the frequency-domain decomposition branch by concatenating it with the current image or feature input X along the channel dimension

$$z = \mathrm{Conv}_{3\times 3}\left([X, P]\right) \in \mathbb{R}^{B\times C\times H\times W} \tag{19}$$

where $[\cdot,\cdot]$ denotes channel-wise concatenation. Based on the conditioned representation z, the network predicts the ellipse parameters Θ from which the adaptive frequency mask M is subsequently constructed. It should be emphasized that PGMAM is related to top-down semantic guidance in a broad sense, but it differs from standard top-down feature feedback in both *representation form* and *functional role*. Conventional top-down pathways directly propagate multichannel high-level feature tensors to lower stages for feature fusion or semantic enrichment. By contrast, PGMAM first compresses decoder semantics into a single-channel normalized prior map and then uses this map to influence downstream processing in a constrained manner. Therefore, its role is not direct semantic supplementation but prior-guided modulation of the subsequent mask generation and feature adaptation process.

The injected prior further influences the adaptive frequency mask in an explicit and differentiable manner. Specifically, since the ellipse parameters Θ are predicted from z through global average pooling, 1×1 convolution, GELU activation, and Sigmoid normalization, the gradient path from the prior map to the ellipse parameters satisfies

$$\frac{\partial \Theta}{\partial P} = \frac{\partial \Theta}{\partial z} \cdot \frac{\partial z}{\partial P} \neq 0. \tag{20}$$

This indicates that the prior map directly participates in the estimation of the adaptive frequency mask M. Consequently, the downstream high-frequency or low-frequency decomposition is conditioned on a coarse category-aware preference rather than being solely determined by the raw input representation.

At the final decoding stage, we further inject the lowest-level prior into the preprediction representation. Specifically, for the lowest decoder feature $U^{(1)} \in \mathbb{R}^{B\times C_d\times H_1\times W_1}$, we introduce a linear projection

$$\begin{aligned} &\Phi : \mathbb{R}^{B\times 1\times H_1\times W_1} \to \mathbb{R}^{B\times C_d\times H_1\times W_1} \\ &\Phi\left(P^{(1)}\right) = W_\Phi * P^{(1)} \end{aligned} \tag{21}$$

where W_Φ denotes a 1×1 convolution. The projected prior is then fused with $U^{(1)}$ through residual addition

$$\tilde{U}^{(1)} = U^{(1)} + \Phi\left(P^{(1)}\right). \tag{22}$$

This residual injection introduces an explicit semantic bias before the segmentation head, thereby enhancing spatial consistency and category discrimination in the final prediction stage. From this perspective, the prior functions as a coarse category-aware preference imposed before final decision making.

Finally, $\tilde{U}^{(1)}$ is fed into the segmentation head f_{seg}, which consists of a 3×3 convolution, batch normalization, ReLU, dropout, and a 1×1 convolution

$$\begin{aligned} \hat{Y} &= f_{seg}\left(\tilde{U}^{(1)}\right) \\ &= Conv_{1\times 1}\left(Dropout\left(ReLU\left(BN\left(Conv_{3\times 3}\left(\tilde{U}^{(1)}\right)\right)\right)\right)\right) \end{aligned} \tag{23}$$

where BN denotes batch normalization.

Overall, PGMAM can be understood as a constrained form of semantic guidance rather than directly feeding back high-level feature tensors for fusion. It distills decoder semantics

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into an internally induced semantic prior and uses this map to guide downstream frequency mask generation and final prediction refinement. Such a design gives the module a clear functional interpretation as prior-guided modulation rather than generic contextual enhancement.

IV. Datasets and Experimental Settings

A. Datasets Description

1) *Vaihingen Dataset:* There exist 33 annotated aerial tiles in this dataset. A total of 15 tiles (IDs: 1, 3, 5, 7, 11, 13, 15, 17, 21, 23, 26, 28, 32, 34, and 37) are used for training, with tile 30 serving as the validation set, while the remaining 17 tiles are used for testing. Pixel-level annotations are provided for five land-cover categories, including car, tree, low vegetation, building, impervious surface, and one class termed clutter.
2) *Potsdam Dataset:* The Potsdam benchmark consists of 38 uniformly sized aerial tiles. Following the experimental protocol, 24 tiles are assigned to the training set (IDs: 2_10, 2_12, 3_11, 3_12, 4_10, 4_11, 5_10, 5_12, 6_8, 6_9, 6_10, 6_11, 6_12, 7_7, 7_9, 7_10, 7_11, and 7_12), while six tiles (IDs: 2_11,3_10, 4_12, 5_11, 6_7, and 7_8.) are reserved for validation. Remaining tiles are used exclusively for testing. The annotation protocol follows the same as that used in the Vaihingen dataset.
3) *LoveDA Dataset [55]:* The LoveDA benchmark is short for LandcOVE dataset for domain adaptation, which advances both semantic segmentation and transfer learning on the remote sensing domain. The LoveDA dataset consists of 5987 images from three different cities. Background, building, road, water, barren, forest, and agriculture are seven labeled categories. While the test sets are not available for researchers, for the LoveDA dataset, the results on the validation set will be reported with all the parameters fixed following [45] and [56].

B. Evaluation Metrics

Following standard evaluation practices, segmentation performance is quantified using class-wise overlap accuracy and F1 measure, together with their aggregated statistics over the entire dataset: mIoU and mean F1. The class-wise overlap accuracy, termed IoU, is a normative performance metric that measures the overlapping degree between the predicted object regions and the substantive object regions specified by the ground-truth annotations. The F1 scores are considered as the combined assessment of precision and recall for a certain category. In addition, OA reflects the overall classification accuracy by summarizing correct predictions over all image pixels. The corresponding metrics are formulated as follows:

$$IoU_k = \frac{TP_k}{TP_k + FP_k + FN_k}, \quad mIoU = \frac{1}{n}\sum_{k=1}^{n} IoU_k \tag{24}$$

$$Precision_k = \frac{TP_k}{TP_k + FP_k}, \quad Recall_k = \frac{TP_k}{TP_k + FN_k} \tag{25}$$

$$F1_k = \frac{2 \cdot Precision_k \cdot Recall_k}{Precision_k + Recall_k}, \quad mF1 = \frac{1}{n}\sum_{k=1}^{n} F1_k \tag{26}$$

$$OA = \frac{\sum_{k=1}^{n} TP_k}{\sum_{k=1}^{n} (TP_k + FP_k + TN_k + FN_k)} \tag{27}$$

where TP_k, FP_k, TN_k, and FN_k indicate the true positive, false positive, true negative, and false negative samples, respectively, for each class k.

C. Implementation Details

ConvNeXt-Tiny is used as the encoder, and all models are optimized with Adam at a learning rate of 6×10^{-4}. Following common practice in RSSS [57], we adopt dataset-specific training budgets to account for different convergence behaviors. Specifically, the maximum number of training epochs is set to 50 for LoveDA and 250 for Vaihingen and Potsdam, respectively. Standard training strategies, including learning rate scheduling, regularization techniques, and validation-based model selection, are adopted to ensure stable and fair evaluation across all datasets and comparison methods [58].

V. Ablation Study and Module Analysis

The framework comprises three core modules, namely MEFDM, CDAFM, and PGMAM, where CDAFM and PGMAM are constructed on top of MEFDM. The baseline is constructed by the encoder ConvNeXt-Tiny and some other decoding stages consisting of basic convolutional transformation. Models incorporating additional components are denoted using the format *baseline* + *module*. For instance, the variant integrating MEFDM is referred to as *baseline+MEFDM* consistent with the notation used in several tables. And the CDAFM and PGMAM are built upon the MEFDM; therefore, the comparison for validating the effectiveness of these two modules is between *baseline+MEFDM* and *baseline+MEFDM+CDAFM*, and *baseline+MEFDM+PGMAM*.

A. Performance Analysis of MEFDM

First, we compare the *baseline* with *baseline+MEFDM*. Table II summarizes the quantitative ablation results, demonstrating that incorporating MEFDM yields performance gains across all evaluation criteria. Concretely, mIoU, mF1, and OA increase from 82.57, 90.23, and 92.72 to 83.67, 90.91, and 93.02, respectively, confirming the overall benefit of this component. At the class level, notable IoU improvements are achieved in all of the categories.

Furthermore, the qualitative results in Fig. 5 illustrate that *baseline+MEFDM* yields more distinct and well-separated object boundaries, while simultaneously strengthening the recognition of targets exhibiting strong directional characteristics. Specifically, in the first row, the three "car" instances are delineated with clearer contours, effectively alleviating boundary adhesion between adjacent objects. In the second row, the large-area "tree" region is more completely and accurately segmented after introducing MEFDM, avoiding the misclassification of boundary pixels as "low vegetation" that occurs in the results obtained from *baseline*. In the last row, the "tree" instance with a near right-angle structure on the left

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TABLE II
ABLATION STUDY RESULTS OF EACH WHOLE MODULE ON VAIHINGEN DATASET

Line Index	Methods	Per-class IoU					Overall Metrics		
		Impervious Surface	Building	Low Vegetation	Tree	Car	mIoU	mF1	OA
1	Baseline	93.36	90.70	71.82	80.40	76.54	82.57	90.23	92.72
2	Baseline+MEFDM	93.55	91.27	72.28	81.35	79.91	83.67	90.91	93.02
3	Baseline+MEFDM+CDAFM	93.88	91.41	72.40	81.37	80.01	83.81	90.99	93.13
4	Baseline+MEFDM+PGMAM	93.88	91.31	72.61	81.54	79.83	83.83	91.01	93.15
5	**PEDNet**	**94.25**	**91.85**	**73.26**	**81.65**	**81.05**	**84.41**	**91.36**	**93.40**

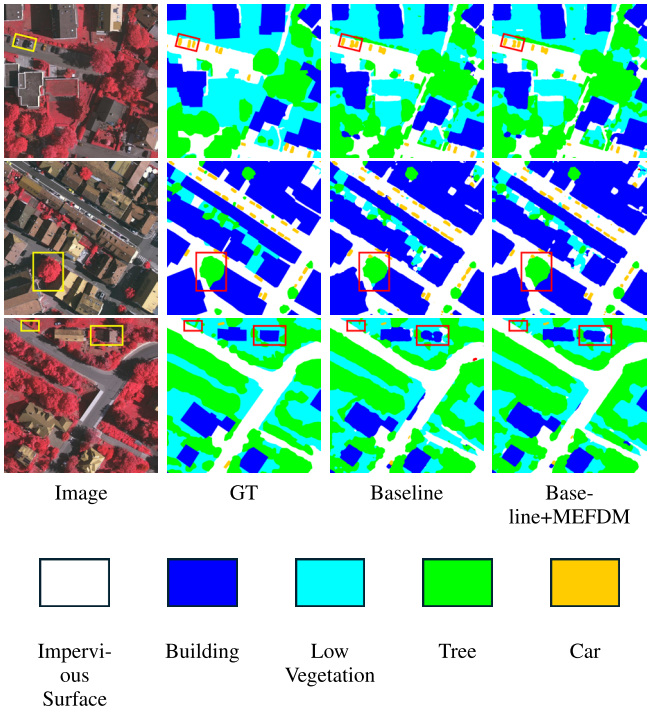


Fig. 5. Visualization results of MEFDM. (Left to right) Images correspond to the original image, the ground-truth map, the result produced by the baseline, and the result produced by the baseline augmented with the MEFDM.

TABLE III
COMPUTATIONAL METRICS RESULTS FOR ABLATION STUDY

Line Index	Methods	Computational Metrics	
		Params(M)	FLOPs(G)
1	Baseline	30.21	26.69
2	Baseline+MEFDM	32.75	27.95
3	Baseline+MEFDM+CDAFM	38.82	31.54
4	Baseline+MEFDM+PGMAM	32.75	27.96
5	**PEDNet**	38.83	31.54

is identified more precisely compared with the baseline, and the "building" on the right no longer suffers from erroneous "low vegetation" predictions within its interior region.

As reported in Table III, incorporating MEFDM incurs an additional computational overhead, yet the resulting cost increase is modest when weighed against the achieved performance gains, indicating a balanced efficiency-accuracy profile.

TABLE IV
ABLATION STUDY OF PERFORMANCE ON DIFFERENT NUMBERS OF ELLIPSES

Line Index	Methods	Overall Metrics		
		mIoU	mF1	OA
1	Baseline	82.57	90.23	92.72
2	Baseline+MEFDM(1)	83.37	90.71	93.01
3	Baseline+MEFDM(2)	83.21	90.63	92.99
4	Baseline+MEFDM(4)	**83.67**	**90.91**	**93.02**
5	Baseline+MEFDM(8)	83.31	90.68	93.02
6	Baseline+MEFDM(16)	83.28	90.67	92.90
7	Baseline+MEFDM(32)	83.16	90.59	92.99

TABLE V
ABLATION STUDY OF COMPUTATIONAL COMPLEXITY ON DIFFERENT NUMBERS OF ELLIPSES

Line Index	Methods	Computational Metrics	
		Params(M)	FLOPs(G)
1	Baseline	30.21	26.69
2	Baseline+MEFDM(1)	32.75	27.95
3	Baseline+MEFDM(2)	32.75	27.95
4	Baseline+MEFDM(4)	32.75	27.95
5	Baseline+MEFDM(8)	32.75	27.95
6	Baseline+MEFDM(16)	32.75	27.95
7	Baseline+MEFDM(32)	32.76	27.95

B. Parameter Sensitivity and Comparative Analysis of MEFDM

Moreover, we further analyze its internal parameters and compare it with related modules to validate its design rationality.

1) Parameter Sensitivity: Table IV analyzes the behavior of MEFDM under varying ellipse configurations, which control the directional sensitivity and structural granularity of frequency decomposition. We examine six candidate settings of the parameter K, the number of ellipses with 1, 2, 4, 8, 16, and 32, to assess their influence on segmentation quality. As shown in Table IV, the setting with *four ellipses* yields the best overall results, although the differences among the tested configurations are relatively small. This observation suggests that increasing the number of ellipses enhances the diversity of local pattern modeling, whereas excessive partitioning offers diminishing returns.

Table V reports the variation in model complexity, including parameter count and computational cost, under different ellipse configurations, indicating that modest increases in the number

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TABLE VI
COMPARISON STUDY FOR THE SHAPE OF FREQUENCY MASK ON PERFORMANCE BETWEEN OURS AND AFENET

Line Index	Methods	Overall Metrics		
		mIoU	mF1	OA
1	Baseline+AFSIM [59]	83.47	90.77	92.98
2	Baseline+MEFDM(4)	**83.67**	**90.91**	**93.02**

TABLE VII
COMPARISON STUDY OF THE SHAPE OF FREQUENCY MASK ON COMPUTATIONAL COMPLEXITY BETWEEN OURS AND AFENET

Line Index	Methods	Computational Metrics	
		Params(M)	FLOPs(G)
1	Baseline+AFSIM [59]	32.73	27.94
2	Baseline+MEFDM(4)	32.75	27.95

TABLE VIII
COMPARISON WITH REPRESENTATIVE FREQUENCY MASKING BASELINES ON THE VAIHINGEN DATASET

Line Index	Methods	Overall Metrics		
		mIoU	mF1	OA
1	Baseline+Adaptive Masking	83.54	90.82	93.11
2	Baseline+Circular Masking	83.18	90.61	92.97
3	Baseline+Learnable Masking	83.64	90.89	**93.13**
4	**Baseline+MEFDM(4)(Ours)**	**83.67**	**90.91**	93.02

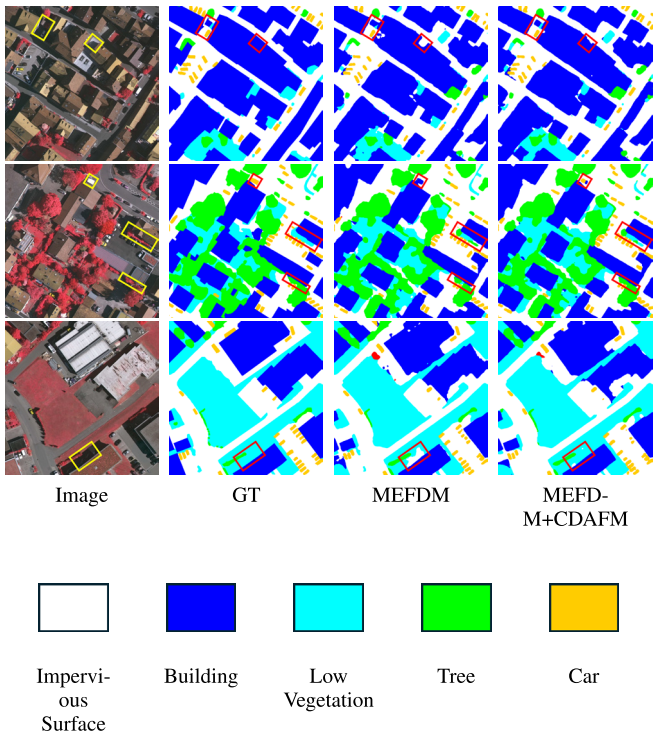


Fig. 6. Visualization results of CDAFM. (Left to right) Images correspond to the original image, the ground-truth map, the result produced by the baseline augmented with MEFDM, and the result produced by the baseline augmented with MEFDM and CDAFM.

of ellipses incur negligible changes in both the parameter size and computational overhead. Based on such observation, the number of ellipses is fixed to 4 in all subsequent experiments.

2) Comparison With Similar Frequency-Decomposition Modules: To verify the effect of MEFDM, we further replace it with a representative counterpart that the AFSIM module from AFENet [59] while keeping the remaining network architecture unchanged. Since both AFSIM and MEFDM are designed for explicit frequency-domain decomposition, this comparison is intended as *a module-level ablation* rather than a full-framework comparison. The results are reported in Table VI. It can be observed that MEFDM achieves higher overall metrics than AFSIM, with notable improvements in mIoU (+0.20), mF1 (+0.14), and OA (+0.04). These results suggest that the proposed multielliptical decomposition strategy provides more discriminative and stable feature representations than the conventional square mask-based counterpart, while introducing only a slight parameter increase, as shown in Table VII.

3) Comparison With Representative Frequency Masking Baselines: We further compare the proposed multielliptical masking with several representative strategies, including adaptive masking, learnable frequency masking, and circular masking under the same experimental setting. As shown in Table VIII, the proposed method achieves the highest mIoU (83.67) and mF1 (90.91), while the learnable mask obtains a slightly higher OA (93.13). These results suggest that the proposed structured multielliptical masking is beneficial for improving region-level segmentation quality and class-balanced performance. In addition, the circular masking strategy performs worse than the proposed method on all three metrics, which may imply that isotropic frequency selection is less suitable for describing the anisotropic and direction-sensitive spectral patterns commonly encountered in RSIs. Overall, these comparisons provide additional empirical support for the effectiveness of the proposed masking design.

C. Performance Analysis of CDAFM

The effectiveness of CDAFM is further examined through ablation experiments. Results summarized in Lines 2 and 3 of Table II show that incorporating CDAFM yields consistent improvements in mIoU from 83.67 to 83.81, mF1 from 90.99 to 91.01, and OA from 93.02 to 93.13, highlighting its role in strengthening feature discrimination and contextual modeling.

The qualitative comparisons are illustrated in Fig. 6. Specifically, in the first row, three small objects located on the left are examined, corresponding to a "car," "building," and "low vegetation." In this case, the result of *baseline+MEFDM* exhibits misclassification of "low vegetation" as "building," while the "building" on the right is fragmented and fails to be recognized as a coherent structure. In contrast, the model augmented with CDAFM correctly distinguishes all three small objects and simultaneously demonstrates the ability to capture spatially continuous, long-range structures. In the second row, the upper example focuses on "car" segmentation, where CDAFM achieves higher accuracy than the *baseline+MEFDM*, which incorrectly labels a tiny "low vegetation" region as "tree," whereas this error is effectively

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TABLE IX
ABLATION STUDY OF DIFFERENT INTERACTION COMPONENTS IN CDAFM ON THE VAIHINGEN DATASET

Line Index	Methods	Overall Metrics		
		mIoU	mF1	OA
1	Window self-attention only	83.70	90.92	93.11
2	Cross-attention only	83.76	90.97	**93.15**
3	Gated fusion only	83.72	90.94	93.07
4	Full CDAFM	**83.81**	**90.99**	93.13

TABLE X
ABLATION STUDY ON PERFORMANCE FOR DIFFERENT APPROACHES FOR FEATURE FUSION

Line Index	Methods	Overall Metrics		
		mIoU	mF1	OA
1	Baseline+MEFDM+Plus	83.70	90.93	93.07
2	Baseline+MEFDM+Cat	83.66	90.90	93.10
3	Baseline+MEFDM+CDAFM	**83.81**	**90.99**	**93.13**

resolved after introducing CDAFM at the lower location of the image. The third row further highlights the long-range coherence modeling capability of CDAFM, where it successfully recovers a continuous "low vegetation" structure that the *baseline* fails to segment consistently.

To clarify the contribution of each interaction operation in CDAFM, we further conduct an isolated ablation study on three key components: window self-attention, cross-attention, and gated fusion as reported in Table IX. It can be observed that each individual component yields measurable improvements, indicating that the performance gain of CDAFM does not arise from a redundant stacking of operations. Among the three single-component variants, the cross-attention configuration achieves the best performance with the mIoU of 83.76, the mF1 of 90.97, and the OA of 93.15. These results suggest that explicit bidirectional interaction between spatial domain and frequency domain features is the primary source of improvement. In comparison, window self-attention and gated fusion also provide consistent gains, showing that intradomain contextual refinement and adaptive information selection are both beneficial. When three components are jointly enabled, the full CDAFM achieves the best overall performance, which demonstrates that these operations are complementary rather than redundant. This result confirms that the effectiveness of CDAFM comes from the synergy of intra-domain refinement, cross-domain interaction, and adaptive fusion.

We further compare CDAFM with two straightforward yet effective fusion mechanisms, namely element-wise addition and feature concatenation, as shown in Table X. As a result, CDAFM achieves the most obvious improvement in accuracy, demonstrating its superior effectiveness.

As shown in Table XI, CDAFM results in a parameter increase from 32.75 to 38.82M and computational complexity from 27.95 to 31.54 GFlops. This additional overhead is accompanied by consistent improvements in segmentation performance, indicating an effective tradeoff between modeling capacity and accuracy.

TABLE XI
ABLATION STUDY ON COMPLEXITY FOR DIFFERENT APPROACHES FOR FEATURE FUSION

Line Index	Methods	Computational Metrics	
		Params(M)	FLOPs(G)
1	Baseline+MEFDM+Plus	32.75	27.95
2	Baseline+MEFDM+Cat	33.52	28.4
3	Baseline+MEFDM+CDAFM	38.82	31.54

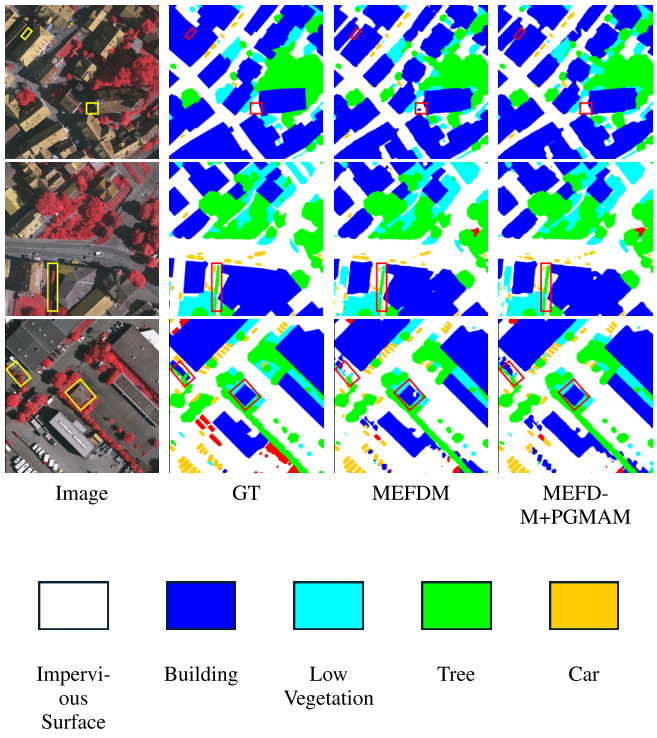


Fig. 7. Visualization results of PGMAM. (Left to right) Images correspond to the original image, the ground-truth map, the result produced by the baseline augmented with MEFDM, and the result produced by the baseline augmented with MEFDM and PGMAM.

D. Performance Analysis of PGMAM

After confirming the effectiveness of MEFDM and CDAFM, we further examine the role of PGMAM, which is designed to enhance context perception by introducing semantic priors. As reported in Table II, adding PGMAM consistently improves performance across all the metrics.

Fig. 7 provides a more intuitive understanding of how PGMAM improves class discrimination by reducing confusion between land-cover categories with similar spectral or structural characteristics. Three representative visualization examples are shown. In the first row, a narrow elongated region corresponding to a "building" is incorrectly classified as "impervious surface" by the *baseline+MEFDM* model, while a nearby "impervious surface" is misidentified as "building." After introducing PGMAM, both regions are correctly identified, indicating improved differentiation between visually

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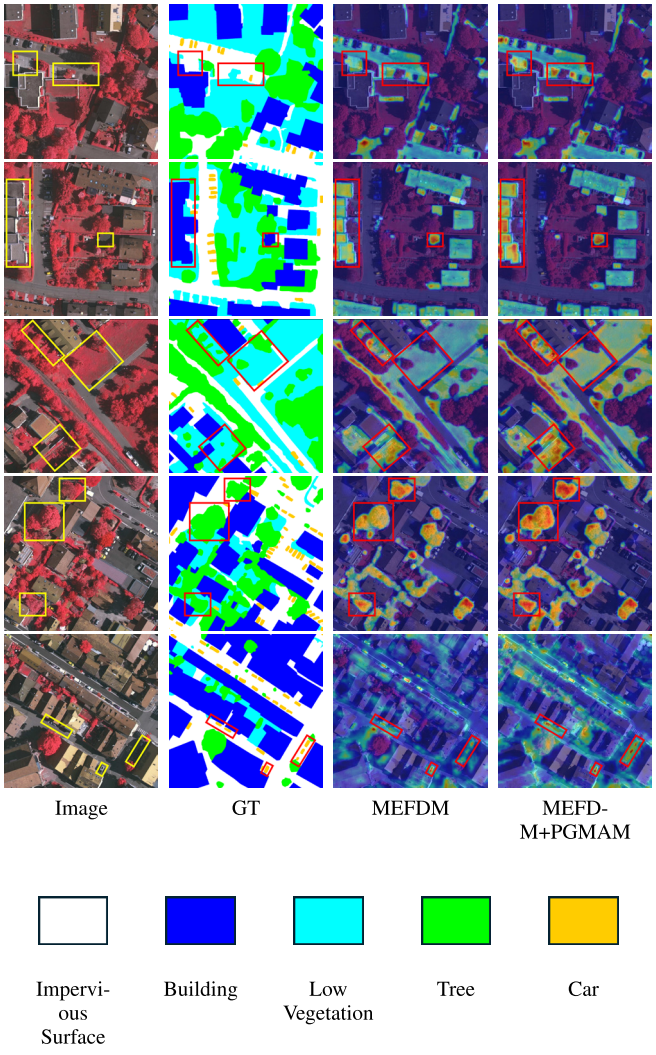


Fig. 8. Visualization results of PGMAM on the attention map. (Left to right) Attention map corresponds to the original image, the ground-truth map, the result produced by the baseline augmented with MEFDM, and the result produced by the baseline augmented with MEFDM and PGMAM.

similar man-made classes. The second row shows that an elongated "tree" region is partially misclassified as "low vegetation" by the *baseline+MEFDM* model. This confusion is effectively resolved once PGMAM is added, indicating clearer semantic separation between vegetation-related categories. In the third row, adjacent "building" and "tree" regions are both misclassified as "impervious surface" and "low vegetation," respectively, in the absence of PGMAM, whereas the model equipped with PGMAM successfully recovers the correct labels, particularly achieving accurate recognition of the "building" region on the right.

These observations demonstrate that PGMAM effectively enhances class-level discriminability by incorporating category-aware priors, which helps suppress misclassification among land-cover types with overlapping appearances.

To further analyze the impact of PGMAM, we visualize attention heatmaps before and after integrating this module, respectively, as shown in Fig. 8. For each category, representative heatmaps are presented, where brighter regions indicate stronger attention responses. Compared with models without PGMAM, the attention maps become more focused on the relevant target regions after introducing the module, suggesting that PGMAM encourages the network to attend to more semantically meaningful areas and improves category-aware feature representation.

TABLE XII

BACKBONE GENERALIZATION ANALYSIS OF PEDNET ON THE VAIHINGEN DATASET. PEDNET CONSISTENTLY IMPROVES THE CORRESPONDING BASELINE UNDER DIFFERENT LIGHTWEIGHT BACKBONE CHOICES, DEMONSTRATING THAT THE PROPOSED DESIGN IS NOT TIED TO A SPECIFIC ENCODER ARCHITECTURE

Line Index	Method	Overall Metrics		
		mIoU	mF1	OA
1	ResNet-18	82.07	89.90	92.45
2	ResNet-18 + PEDNet	83.05	90.53	92.83
3	Gain	**+0.98**	**+0.63**	**+0.38**
4	Swin-T	81.26	89.38	92.38
5	Swin-T + PEDNet	82.97	90.49	92.84
6	Gain	**+1.71**	**+1.11**	**+0.46**

E. Backbone Generalization Analysis

The above ablation results verify the effectiveness of each individual module from a component-level perspective. Beyond these module-wise analyses, it is also necessary to examine whether the performance gain of the complete PEDNet depends on a specific encoder choice. To this end, we further conduct a backbone generalization study to evaluate the robustness of the proposed framework under different lightweight backbone architectures.

Specifically, since ConvNeXt-Tiny has been adopted as the default encoder in the original implementation, we additionally replace it with two alternative lightweight backbones, namely ResNet-18 and Swin-T, on the Vaihingen dataset. These two backbones are selected because they are of comparable scale to ConvNeXt-Tiny while representing CNN families and Transformer families, respectively. For each backbone, we compare the corresponding baseline with their PEDNet-enhanced version under the same experimental setting.

As reported in Table XII, PEDNet consistently improves the baseline performance under both backbone settings. Specifically, when ResNet-18 is used as the encoder, PEDNet increases mIoU from 82.07 to 83.05, mF1 from 89.90 to 90.53, and OA from 92.45 to 92.83. Similarly, when Swin-T is adopted, PEDNet improves mIoU from 81.26 to 82.97, mF1 from 89.38 to 90.49, and OA from 92.38 to 92.84. These consistent gains indicate that the proposed framework is not tied to a particular backbone architecture, but can generalize well across different encoder families.

It is also observed that the absolute performance of Swin-T is slightly lower than that of ResNet-18 in this experiment. This suggests that the effectiveness of a backbone may vary with the dataset characteristics and experimental setting. Nevertheless, PEDNet consistently improves the corresponding baseline in both cases, which further supports the robustness

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TABLE XIII
COMPARISON RESULTS WITH SOTA METHODS ON THE VAIHINGEN DATASET

Methods	Per-class IoU					Overall Metrics		
	Impervious Surface	Building	Low Vegetation	Tree	Car	mIoU	mF1	OA
FCN(2015) [14]	91.61	85.77	67.90	78.57	60.24	76.82	86.41	91.05
UNet(2015) [15]	89.56	83.46	64.95	77.36	44.14	71.89	82.54	89.71
SegNet(2017) [60]	90.46	83.66	62.91	75.10	41.46	70.72	81.55	89.40
PSPNet(2017) [61]	92.16	87.55	71.94	80.41	62.14	78.84	87.75	92.02
MACUNet(2022) [62]	91.77	86.22	69.97	79.58	56.66	76.84	86.32	91.41
MAResUNet(2022) [63]	93.27	90.44	71.70	81.03	69.40	81.17	89.29	92.66
MANet(2022) [64]	92.69	88.80	70.85	80.63	72.62	81.12	89.33	92.26
SegFormer(2021) [65]	87.83	77.16	60.26	72.33	30.04	65.53	77.20	87.24
UNetFormer(2022) [42]	93.14	90.17	70.67	80.43	76.16	82.11	89.94	92.50
EIGNet(2023) [66]	92.89	90.00	72.16	80.69	75.85	82.32	90.09	92.47
CGGLNet(2024) [67]	93.78	91.49	72.22	81.30	80.49	83.86	91.02	93.13
CMLFormer(2024) [68]	93.08	90.39	71.12	80.52	74.33	81.89	89.80	92.60
SFFNet(2024) [69]	93.82	91.52	73.04	81.56	79.72	83.93	91.07	93.23
PyramidMamba(2025) [70]	93.45	91.15	71.01	80.58	76.39	82.52	90.18	92.77
PPMamba(2025) [71]	93.62	91.17	71.36	79.69	76.96	82.56	90.21	92.75
AFENet(2025) [59]	93.69	91.68	72.72	81.60	80.55	84.05	91.14	93.20
PEDNet	**94.25**	**91.85**	**73.26**	**81.65**	**81.05**	**84.41**	**91.36**	**93.40**

and generalization ability of the proposed method with respect to encoder choice.

VI. COMPARISON WITH SOTA METHODS

PEDNet is evaluated against a broad range of established RSSS methods across three benchmarks. Our comparison set covers both classical segmentation approaches and recent SOTA methods, including FCN [14], UNet [15], SegNet [60], PSPNet [61], MACUNet [62], MAResUNet [63], MANet [64], SegFormer [65], UNetFormer [42], EIGNet [66], CGGLNet [67], CMLFormer [68], PyramidMamba [70], PPMamba [71], AFENet [59], and SFFNet [69].

Among the compared methods, FCN, UNet, SegNet, PSPNet, and SegFormer are general segmentation models originally developed for natural image analysis, while the remaining methods are tailored for RSSS and reflect recent advances in RSSS. MACUNet, MAResUNet, MANet, EIGNet, and CGGLNet are CNN-based methods with different emphases on feature representation, attention modeling, edge perception, and contextual interaction. In addition, UNetFormer and CMLFormer are representative Transformer-based RSSS methods, while PyramidMamba and PPMamba are recent Mamba-based approaches.

Besides the module-level comparison with AFSIM [59] in the ablation study, we further include two representative frequency-aware baselines in the full-model comparison, named AFENet [59] and SFFNet [69]. These two methods reflect different frequency modeling paradigms where AFENet relies on explicit masking-based decomposition, and SFFNet adopts a wavelet-based multiband decomposition strategy. By comparing PEDNet against both methods, we aim to verify that the effectiveness of the proposed framework is not limited to a specific mask-shape design, but generalizes to broader frequency-aware segmentation settings.

All competing methods are evaluated under identical experimental protocols to ensure fair comparison.

A. Results of Vaihingen Dataset

As reported in Table XIII, PEDNet achieves the strongest overall performance on the Vaihingen dataset, attaining the highest mIoU of 84.41, mF1 of 91.36, and OA of 93.40 among all compared methods. These results indicate that PEDNet delivers consistent improvements at both the category level and the dataset level.

From a per-class perspective, PEDNet shows notable gains across all land-cover categories, with particularly pronounced improvements on challenging classes such as “car,” “low vegetation,” and “tree.” Specifically, PEDNet achieves IoU score of 81.05 for “car,” 73.26 for “low vegetation,” and 81.65 for “tree,” outperforming recent strong competitors including AFENet and CGGLNet.

To further complement the quantitative comparison, qualitative visualizations are provided. As shown in the first row of Fig. 9, three “low vegetation” instances are highlighted, where most competing methods partially confuse “low vegetation” with “tree.” In contrast, PEDNet consistently assigns all three regions to the correct category, indicating improved discrimination among vegetation-related classes. The second and third rows focus on “building” recognition and further demonstrate the robustness of PEDNet in resolving class ambiguity. In the second row, several methods misclassify “building” regions as “impervious surface,” whereas PEDNet successfully corrects this error and produces coherent “building” predictions. Similarly, in the third row, a right-angled “building” structure that is incorrectly labeled as “low vegetation” by other approaches is accurately recognized by PEDNet.

Overall, these visual comparisons align well with the quantitative trends observed in Table XIII, confirming that PEDNet

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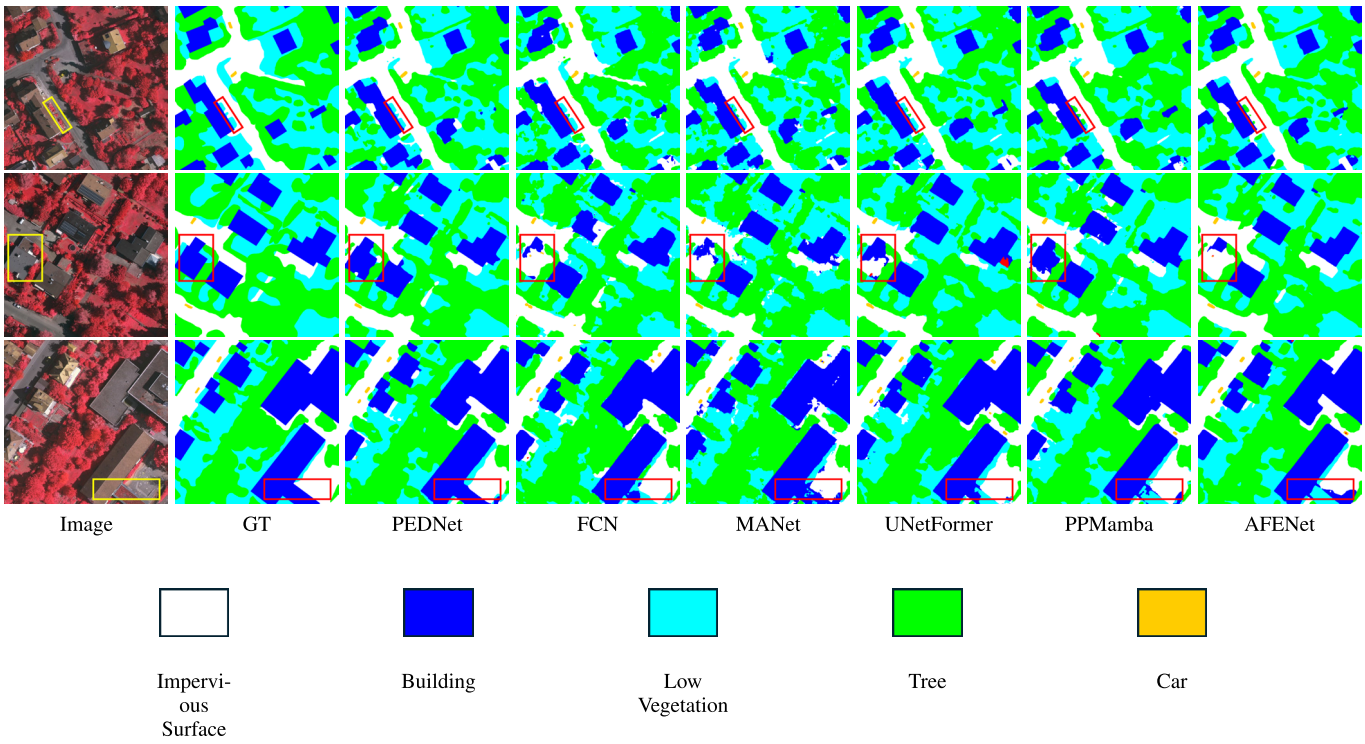


Fig. 9. Visualization results on Vaihingen dataset. (Left to right) Visualization results come from the original image, the ground truth, proposed PEDNet, FCN, MANet, UNetFormer, PPMamba, and AFENet.

TABLE XIV
COMPARISON RESULTS WITH SOTA METHODS ON THE POTSDAM DATASET

Methods	Per-class IoU					Overall Metrics		
	Impervious Surface	Building	Low Vegetation	Tree	Car	mIoU	mF1	OA
FCN(2015) [14]	83.86	86.29	72.10	72.95	86.22	80.28	88.92	87.41
UNet(2015) [15]	84.79	85.99	72.03	73.83	86.96	80.72	89.19	87.66
SegNet(2017) [60]	82.72	86.41	68.73	68.29	81.42	77.51	87.13	86.04
PSPNet(2017) [61]	86.01	89.27	73.23	72.79	86.44	81.55	89.67	88.62
MACUNet(2022) [62]	84.26	86.81	72.68	73.72	85.46	80.59	89.12	87.91
MAResUNet(2022) [63]	88.53	92.84	75.67	76.79	89.94	84.75	91.58	90.56
MANet(2022) [64]	86.62	90.71	75.87	78.56	89.48	84.25	91.34	90.12
SegFormer(2021) [65]	80.46	83.20	66.03	60.44	74.81	72.99	84.09	83.75
UNetFormer(2022) [42]	88.57	92.73	75.35	77.92	92.16	85.35	91.92	90.68
EIGNet(2023) [66]	88.25	92.19	75.75	78.56	92.82	85.51	92.03	90.67
CGGLNet(2024) [67]	89.24	93.37	76.95	79.86	92.95	86.47	92.60	91.38
CMLFormer(2024) [68]	88.73	92.87	75.92	78.59	90.75	85.37	91.96	90.86
SFFNet(2024) [69]	89.27	93.66	77.49	79.79	92.37	86.52	92.63	91.48
PyramidMamba(2025) [70]	87.93	92.47	75.16	78.03	91.99	85.12	91.79	90.51
PPMamba(2025) [71]	89.15	93.26	76.35	78.76	92.21	85.95	92.29	91.17
AFENet(2025) [59]	89.33	93.73	77.39	79.17	93.05	86.53	92.63	91.42
PEDNet	**89.67**	**93.89**	**77.78**	**80.22**	**93.82**	**87.07**	**92.95**	**91.68**

effectively mitigates category confusion among spectrally or structurally similar land-cover types at the algorithmic level.

B. Results of Potsdam Dataset

Table XIV summarizes the quantitative results on the Potsdam dataset. Overall, PEDNet achieves the best performance among all compared methods, with an mIoU of 87.07, an mF1 of 92.95, and an OA of 91.68. These results indicate that PEDNet delivers consistent and reliable improvements at the dataset level.

With regard to the individual class, PEDNet shows stable gains across all five land-cover categories. In particular, the IoU scores for "low vegetation," "tree," and "impervious surface" reach 77.78, 80.22, and 89.67, respectively. These categories are well known to be difficult to separate in

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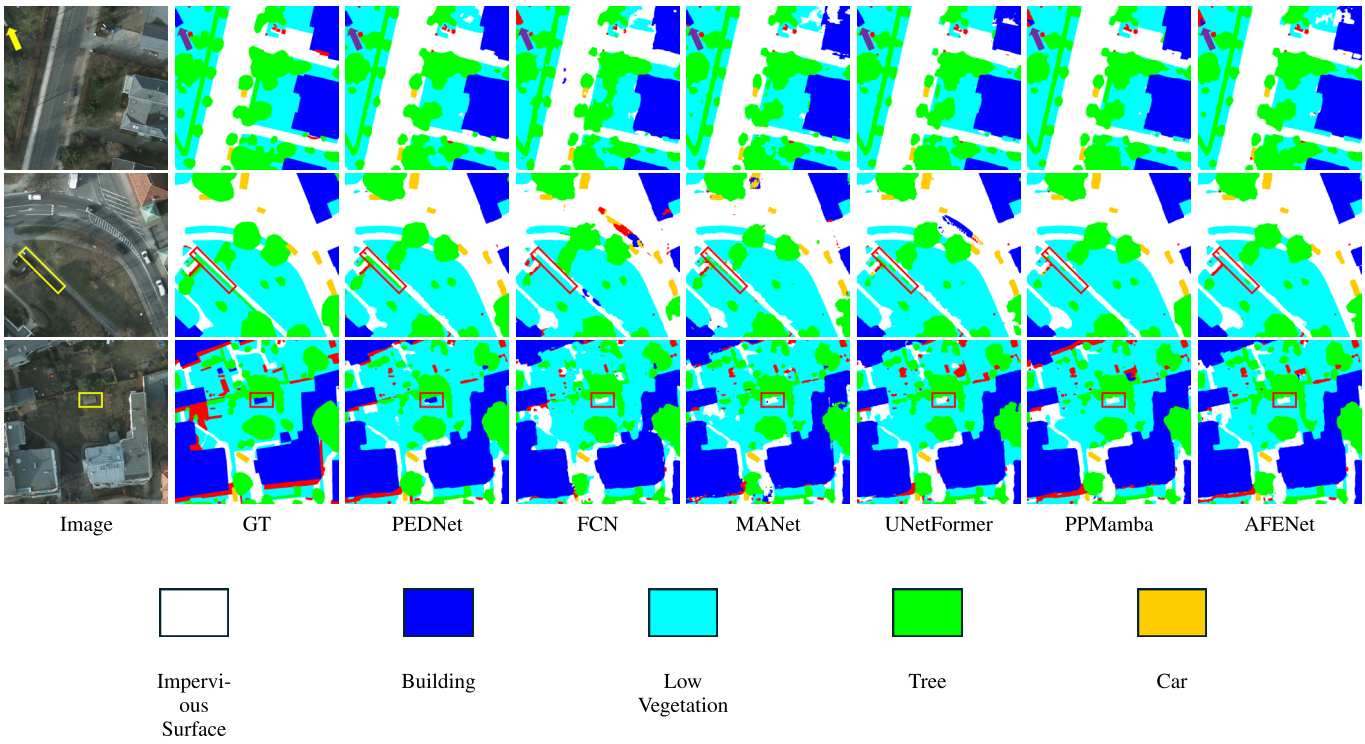


Fig. 10. Visualization results on Potsdam dataset. (Left to right) Visualization results come from the original image, the ground truth, proposed PEDNet, FCN, MANet, UNetFormer, PPMamba, and AFENet.

high-resolution urban scenes due to their similar appearances, and the improved results suggest that PEDNet is better at handling such semantic ambiguity.

The balanced performance on both vegetation-related classes and man-made structures further indicates that PEDNet remains robust under complex urban layouts, where object scales, shapes, and surface materials vary significantly.

In addition to the quantitative results, we provide qualitative comparisons on the Potsdam dataset in Fig. 10 to offer a more intuitive view of PEDNet's behavior. In the first example, a small "building" located near the image boundary is highlighted. While most competing methods mistakenly classify this region as "background" or "clutter," PEDNet correctly identifies it as a "building," demonstrating improved sensitivity to small and boundary adjacent objects. The second row focuses on a slender and elongated "tree" structure. Other methods tend to produce fragmented or broken predictions along this region, whereas PEDNet generates a more complete and coherent segmentation, showing its ability to preserve structural continuity for objects with strong directional patterns. The third example further illustrates PEDNet's robustness in complex contexts. In the upper case, a "building" surrounded by "tree" and "low vegetation" is correctly recognized by PEDNet, while other approaches often confuse it with "impervious surface."

C. Results of LoveDA Dataset

Table XV reports the quantitative comparison results on the LoveDA dataset. Overall, PEDNet achieves the best performance in terms of mIoU, mF1, and OA, reaching 55.87, 70.98, and 73.57, respectively. Although PEDNet ranks second on the "road," its consistently strong overall performance highlights its solid generalization ability across large-scale scenes with diverse land-cover distributions.

From a class-wise perspective, PEDNet shows notable improvements on several challenging categories. Specifically, the IoU scores for "building," "water," "barren," "forest," and "agriculture" reach 64.37, 72.63, 35.70, 45.59, and 61.64, respectively. These classes typically exhibit high intra-class variation and complex spatial layouts across different regions, making them difficult to segment accurately. The observed gains indicate that PEDNet handles such variability more effectively than competing methods.

Moreover, qualitative visualizations on the LoveDA dataset are presented for comparison in Fig. 11. In the first row, a "forest" region located between two "water" bodies is highlighted. Most competing methods tend to overextend "forest" predictions into adjacent areas, leading to the misclassification of "background" or "agriculture." In contrast, PEDNet produces a segmentation result that aligns more closely with the ground truth, demonstrating better category discrimination in areas with strong cross-class adjacency. The second row focuses on a "water" region surrounded by "forest" areas. While other methods generate fragmented or partially incorrect predictions by confusing "water" with nearby categories such as "forest" or "road," PEDNet yields a more continuous and coherent delineation of the "water" body. The third row further presents an even more challenging case, where multiple "water" regions are separated by "background" areas. Only PEDNet successfully preserves all three "water" segments,

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TABLE XV
COMPARISON RESULTS WITH SOTA METHODS ON THE LOVEDA DATASET

Methods	Per-class IoU							Overall Metrics		
	Background	Building	Road	Water	Barren	Forest	Agriculture	mIoU	mF1	OA
FCN(2015) [14]	48.60	51.38	44.54	47.31	24.07	39.32	45.05	42.90	59.50	62.96
UNet(2015) [15]	50.70	51.54	44.00	45.71	24.93	39.77	45.72	43.20	59.82	63.83
SegNet(2017) [60]	47.92	49.48	41.09	41.65	23.07	39.56	41.79	40.65	57.31	61.01
PSPNet(2017) [61]	51.02	56.78	44.73	61.94	20.38	40.66	45.86	45.91	61.84	65.95
MACUNet(2022) [62]	50.55	54.81	45.30	50.66	23.59	39.46	43.83	44.03	60.47	64.12
MAResUNet(2022) [63]	52.15	61.15	45.54	54.67	30.29	43.43	49.13	48.05	64.38	67.20
MANet(2022) [64]	52.57	57.61	50.00	64.54	25.34	37.68	51.40	48.45	64.32	68.28
SegFormer(2021) [65]	46.64	33.91	35.63	49.26	16.11	33.08	41.24	36.55	52.67	59.01
UNetFormer(2022) [42]	52.52	55.15	52.07	65.88	31.71	40.70	58.29	50.90	66.79	69.87
EIGNet(2023) [66]	53.20	62.64	52.01	65.42	28.65	42.09	49.25	50.46	66.25	68.78
CGGLNet(2024) [67]	54.51	63.38	54.18	68.17	33.02	43.49	54.57	53.05	68.62	70.67
CMLFormer(2024) [68]	52.11	58.86	53.31	66.58	25.17	40.33	46.70	49.01	64.78	67.57
SFFNet(2024) [69]	54.19	63.34	53.52	69.06	33.34	45.09	60.82	54.19	69.58	72.21
PyramidMamba(2025) [70]	53.61	61.52	51.03	64.56	25.88	37.48	51.66	49.39	65.11	68.66
PPMamba(2025) [71]	52.51	62.89	52.59	69.38	30.16	41.85	48.93	51.19	66.86	68.81
AFENet(2025) [59]	55.28	60.03	**56.11**	70.66	33.98	37.30	52.75	52.30	67.86	70.47
PEDNet	**56.51**	**64.37**	54.65	**72.63**	**35.7**	**45.59**	**61.64**	**55.87**	**70.98**	**73.57**

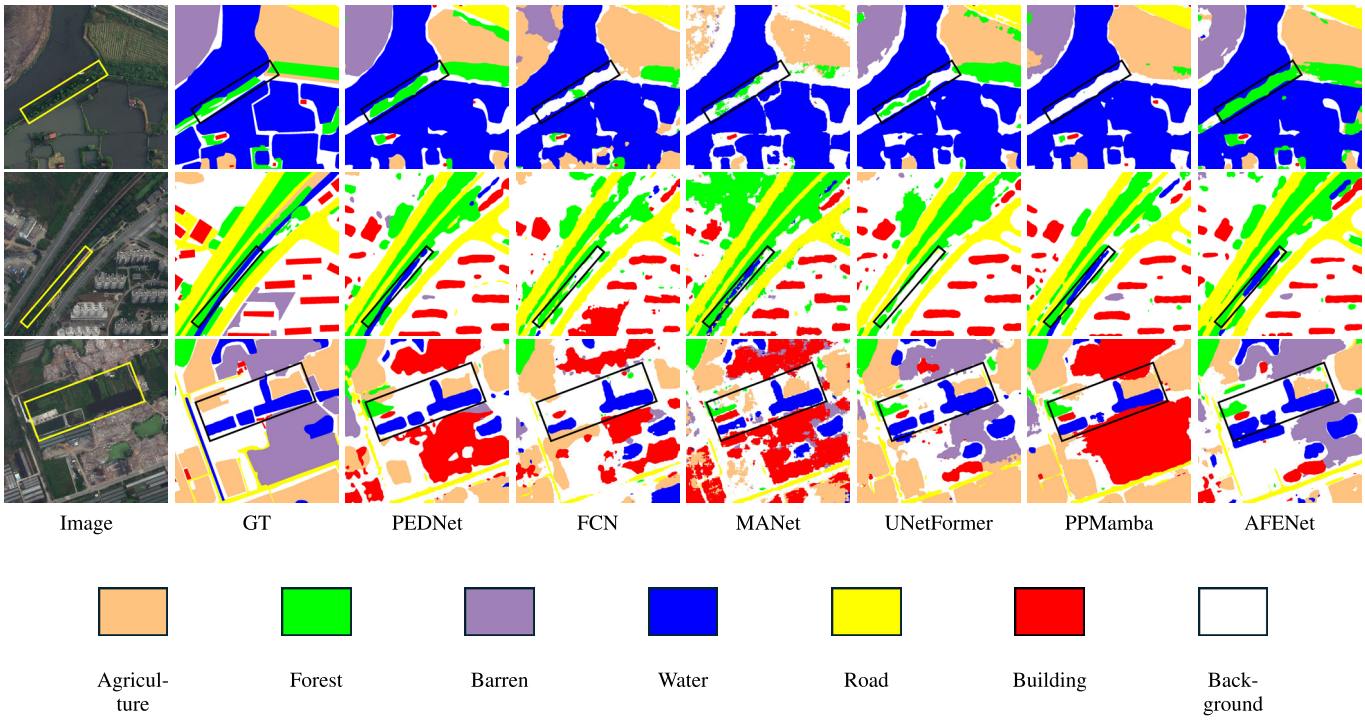


Fig. 11. Visualization results on LoveDA dataset. (Left to right) Visualization results come from the original image, the ground truth, proposed PEDNet, FCN, MANet, UNetFormer, PPMamba, and AFENet.

whereas other approaches show noticeable discontinuities or category confusion.

Overall, these qualitative observations are consistent with the quantitative improvements reported in Table XV. Together, they demonstrate that PEDNet effectively enhances category discrimination and structural consistency across diverse land-cover types and complex scene layouts.

D. Model Parameters and Computation Complexity Analysis

Table XVI summarizes the model size and computational cost of different methods. Compared with lighter baselines, PEDNet incurs a higher computational cost in terms of both parameter count and FLOPs.

This additional overhead is mainly introduced by the frequency-aware decomposition and cross-domain interaction design. Accordingly, PEDNet should be regarded as

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TABLE XVI
COMPARISON WITH SOTA METHODS IN TERMS OF PARAMETER SIZE AND COMPUTATIONAL COMPLEXITY

Methods	Params (M)	FLOPs (G)
FCN [14]	33.63	85.03
UNet [15]	7.77	55.03
SegNet [60]	29.45	161.28
PSPNet [61]	53.32	201.60
MACUNet [62]	5.15	33.61
MAResUNet [63]	26.28	35.11
MANet [64]	35.86	77.76
SegFormer [65]	3.72	6.79
UNetFormer [42]	11.68	11.74
EIGNet [66]	42.13	79.98
CGGLNet [67]	58.34	152.08
CMLFormer [68]	11.81	14.17
CMUNet [72]	13.55	12.67
PyramidMamba [70]	34.42	21.24
PPMamba [71]	30.24	49.18
AFENet [59]	25.59	20.23
PEDNet	**38.83**	**31.54**

an accuracy-oriented framework rather than a lightweight solution, achieving stronger segmentation performance at the expense of increased computation.

VII. CONCLUSION

This work presents PEDNet, a novel framework designed to overcome the limitations of purely spatial-domain modeling in RSSS. Specifically, the MEFDM enables anisotropic and direction-aware frequency modeling, allowing spectral representations to better conform to the geometric characteristics of land-cover structures. The CDAFM further facilitates effective interaction between frequency-domain cues and spatial semantics through bidirectional attention and gated fusion. In addition, the PGMAM dynamically adjusts frequency boundaries using land-cover priors, enhancing category discrimination and semantic consistency across diverse geographical regions. Comprehensive evaluations on three datasets demonstrate the superiority of PEDNet. At the same time, the improved accuracy comes with increased computational cost, which reflects the accuracy-oriented nature of the proposed design rather than a lightweight objective. Future work will focus on reducing the computational burden of the proposed frequency-aware and cross-domain modeling framework while preserving its segmentation advantages on high-resolution RSI.

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